

TED UNIVERSITY

CMPE 491 (Senior Project I)

PROJECT PROPOSAL

Project Name: ALPAR

(Agent Based Limit Aware Planning and Adaptive Routing)

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Project Topic

Safe Route Planning and Route Justification for Mission-Critical Systems

Project Description

Mission-critical systems such as unmanned aerial vehicles (UAVs) and unmanned ground vehicles (UGVs) must operate safely under uncertain and dynamic field conditions. Classical route planning algorithms rely on a static approach: they compute a route once and fall short when conditions change, for example when new obstacles appear, communication is lost, or budget and energy constraints tighten. The aim of this project is to develop an agentic AI based decision and planning architecture that overcomes this limitation by performing instant and safe replanning whenever field conditions change, and to deliver it as a domestically developed software solution.

The project is expected to contribute to three areas. First, it will provide real-time replanning in dynamic crisis scenarios such as increased risk exposure or no-go zone violations. Second, it will include an explainable AI (XAI) decision support module that uses a language model to report why the selected route was preferred over its alternatives, increasing operator trust in the system. Third, it will manage critical system constraints through a dynamic, multi-criteria weighting infrastructure based on a safe energy reserve and uninterrupted communication quality. The architecture will be evaluated in a simulation environment against realistic crisis scenarios. The project is carried out within the SAYZEK-ATP 2026–2027 program, which provides technical and academic mentorship as well as support for testing and validation.

The URL of the project web page

https://berenozbek.github.io/ALPAR_WebPage/